Week 3

Phonological processes

internal use only

Phonemes vs. Allophones

Phonemes

- A phoneme is the **smallest segment of sound** which can **distinguish two words**.
- This is the smallest amount by which these two words could differ and still remain distinct forms. Any smaller subdivision would be impossible because English doesn't subdivide /p/ or /b/. Therefore, /p/ and /b/ are considered two phonemes.

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Minimal pairs

- Pair of words such as 'pit' and 'bit', 'pit' and 'pet', 'back' and 'bag' which differ by only one phoneme **in identical environment** are known as minimal pairs.
- One way to identify the phonemes of any language is to look for minimal pairs.

Phonemes

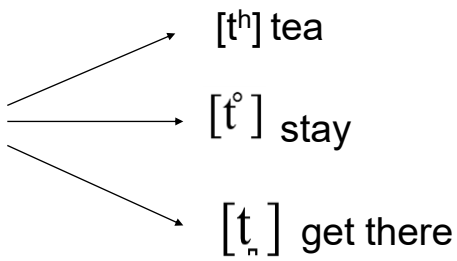
- There are 44 phonemes in English: 24 consonants and 20 vowels.
- Each phoneme is meaningless in isolation. It becomes meaningful only when it is combined with other phonemes.
- Problem:
  - A letter can be represented by different sounds.
  - A phoneme can be represented by different letters or combinations of letters.

=> Phonemes form a set of abstract units that can be used for noting down the sound of a language systemmatically and unambiguously.

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Allophones

- Allophones are the variants of phonemes that occur in speech.
- Reasons: the way a phoneme is pronounced is conditioned by the sounds around it or by its position in the word. For example: /t/



Phonemes vs. Allophones

- Substituting one phoneme for another will result in a word with a different meaning (that's why phonemes can be defined as meaning-distinguishing sounds) as well as a different pronunciation.

  E.g.
- Substituting allophones only results in a different pronunciation of the same words.

  E.g.

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Symbols

Phonemic symbols

- Symbols for phonemes.
- The number of phonemic symbols must be exactly the same as the number of phonemes we decide to exist in the language.
- In RP (BBC English), there are 44 phonemic symbols.

Phonetic symbols

- Symbols for allophones.
- They are used to give an accurate label to an allophone of a phoneme or to represent sounds more accurately.
- Phonetic symbols usually make use of diacritics ([°, ˜ , ʔ , ʰ]).

Phonemic / broad transcription

- A phonemic transcription is a transcription in which each phoneme is represented by one phonemic symbol.
- In other words, in a phonemic transcription, every speech sound must be identified as one of the phonemes and written down with an appropriate symbol.

E.g.:

/wen aɪ lʊk bæk ɒn maɪ tʃaɪldhʊd/

internal use only

Phonetic / narrow transcription

- A phonetic transcription is a transcription which contains a lot of information about the exact quality of the sounds.
- It shows more phonetic detail such as aspiration, length, nasalisation ..., by using a wide variety of symbols and in many cases diacritics.
- E.g: [n̪ ðə tʰp̚ ə ðə wɜːɫd]
- In a phonetic transcription, the symbols are used to represent precise phonetic values, not just to represent phonemes.